AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-30 | Entry window 30 October 2026 (full day) | 24 records screened in scope

24	9	3	1	5
RECORDS IN SCOPE (67 EXCLUDED)	OF 10 SUBTOPIC BANDS POPULATED	EFFECT ESTIMATES WITH A CI	TIER-A RECORDS	RECORDS CARRIED ON METADATA ALONE

Signal of the day

All of Us, 179,218 adults, preregistered: heat anomaly and social vulnerability predict incident depression; 30-day wildfire-smoke $PM_{2.5}$ does not — at 3-digit-ZIP resolution.

Cox models on 5,484 first EHR-recorded depression diagnoses (mean follow-up 371 days). Heat anomaly HR 1.06 (1.03–1.10) and 1.08 (1.04–1.11) per SD; social-material vulnerability HR 1.36 (1.33–1.40) per SD. Mean smoke $PM_{2.5}$ (Stanford ECHO): no association in the primary model; secondary smoke metrics heterogeneous by metric and follow-up window. **Where it is weak:** exposure is a ZIP3 average frozen at baseline — smoke episodes are short and spatially patchy, so this is the textbook setting for attenuation to the null. A preregistered null is still a null on a heavily smoothed exposure, not evidence of no effect. **Why it matters here:** the contrast between a clean heat signal and a flat smoke signal is what residence-scale, humidity-corrected sensor $PM_{2.5}$ would test directly.

What changed today

- **Low-cost sensors as the reference.** A MethodsX protocol measures air-cleaner CADR from $PM_{2.5}$ decay on low-cost sensors in a real home: ∼26 % below lab values from research-grade instruments — conservative, and quantified. A four-filter Corsi–Rosenthal box led the DIY builds.
- **Hygroscopicity is source-dependent.** Eastern China, humidified nephelometry: κ_{OA} 0–0.49 (mean 0.11), tracking biomass-burning influence rather than O:C. A fixed-κ humidity correction is wrong during smoke. Separately, Kelvin-suppressed growth of 15–36 nm particles changes modelled lung deposition by up to 25 %.
- **Policy co-benefit, modelled.** China's 2017 climate-adaptive city pilot: $PM_{2.5}$ $-3.87\,\mu g\,m^{-3}$ (-6.49 to -1.25) by difference-in-differences, ∼8 % lower GEMM burden — shrinking in the matched sample.
- **Prenatal exposure, small signals.** TTN in Taiwan: $PM_{2.5}$ aOR 1.001 (1.000–1.002) per unit, dominated by CO; ECHO cord-blood methylation classifies high $PM_{2.5}$ exposure with AUC only 0.64–0.68.
- **Five metadata-only records (21 %)**, two of them squarely in scope for sensing: PN-PTI counters on gasoline exhaust (*J. Aerosol Sci.*) and e-cigarette aerosol decay. Retry first.

Window continuity

The 29 October issue closed with `last_entry_date` at 29 October. A 31 October 05:33 UTC firing harvested the PubMed and Europe PMC legs for 30 October and stopped. This issue was built on the **1 November 05:25 UTC (00:25 CDT) run**; by the 22:00 rule the newest buildable day is 31 October, so the run backfills **30 October in full** here and 31 October as its own issue, with the W44 weekly (25–31 October) and the October monthly. Candidates were screened sequentially across the two days (cross-day DOI and title set built before authoring) to avoid the double admissions of the 23–29 October backfill.

Corpus analytics

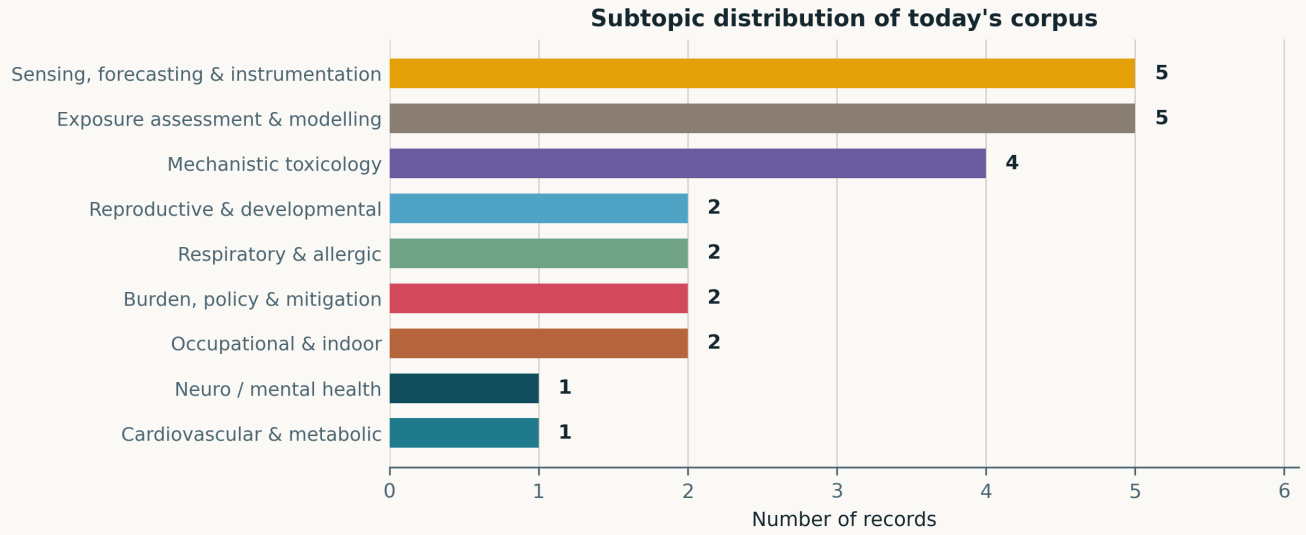


Fig. Subtopic assignment of the 24 in-scope records across nine bands; other clinical endpoints is empty. Sensing and exposure assessment lead with five each, mechanistic toxicology four; reproductive, respiratory, burden and occupational/indoor two each; neuro and cardiovascular one each.

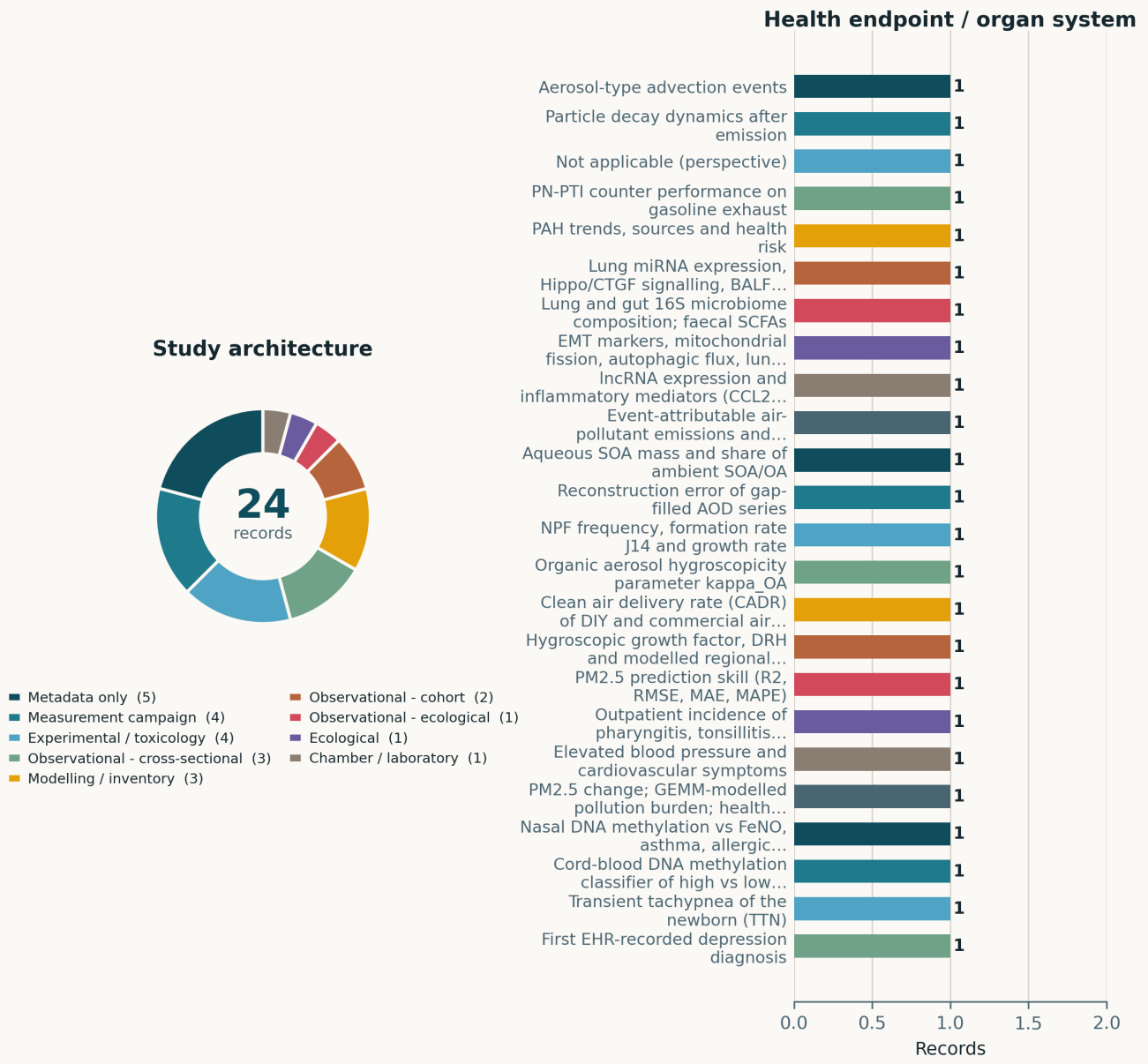


Fig. Left — architecture: **metadata-only 5 (21 %)**, four measurement campaigns, four toxicology, three modelling, three cross-sectional, two cohort-type (All of Us, TTN case-cohort), two ecological and one laboratory (HTDMA). Right — endpoints span depression, TTN, cord-blood methylation, air-cleaner CADR, hygroscopicity and new particle formation.

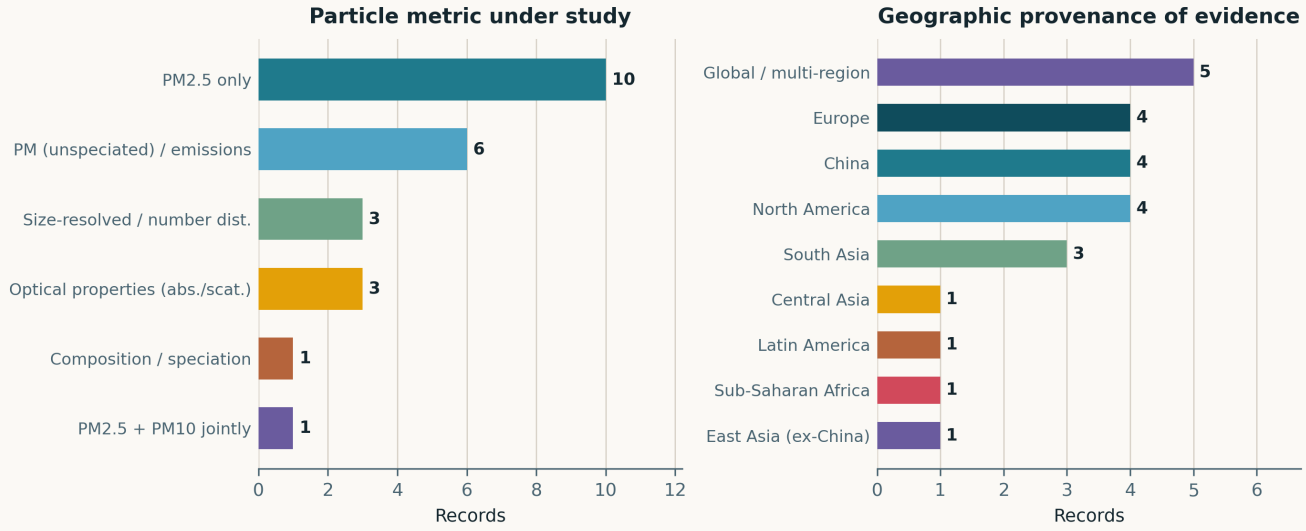


Fig. Left — particle metric: $PM_{2.5}$-only 10, unspeciated/emissions 6, size-resolved/number 3, optical 3, and one each joint $PM_{2.5}$+PM_{10} and composition. Right — geography: global/not stated 5, North America 4, China 4, Europe 4, South Asia 3, and one each East Asia (Taiwan), Sub-Saharan Africa (Tanzania), Latin America (Brazil) and Central Asia (Kazakhstan).

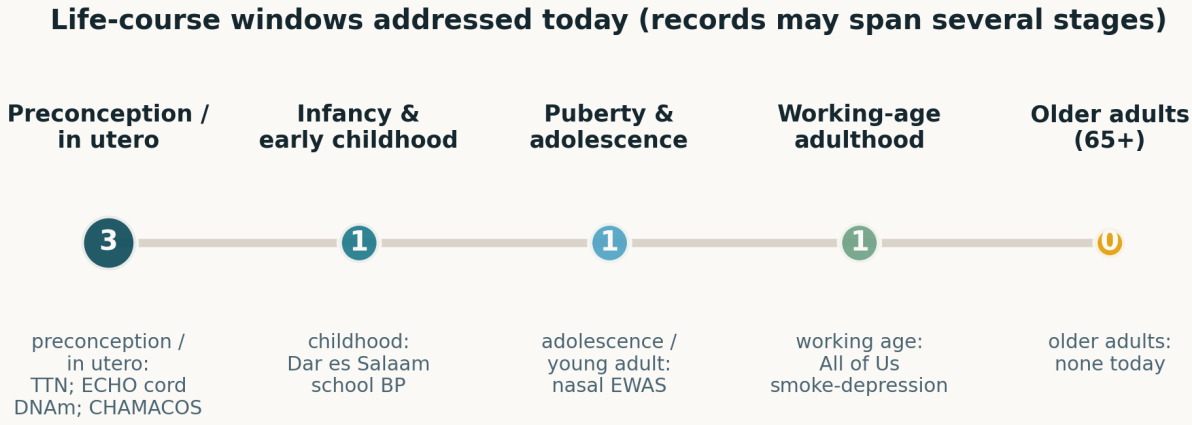


Fig. Life-course windows: three in-utero/early-life records (TTN, ECHO cord blood, CHAMACOS prenatal exposure), one childhood (Dar es Salaam school BP), one adolescent/young-adult (nasal EWAS), one working-age (All of Us depression); no older-adult record today.

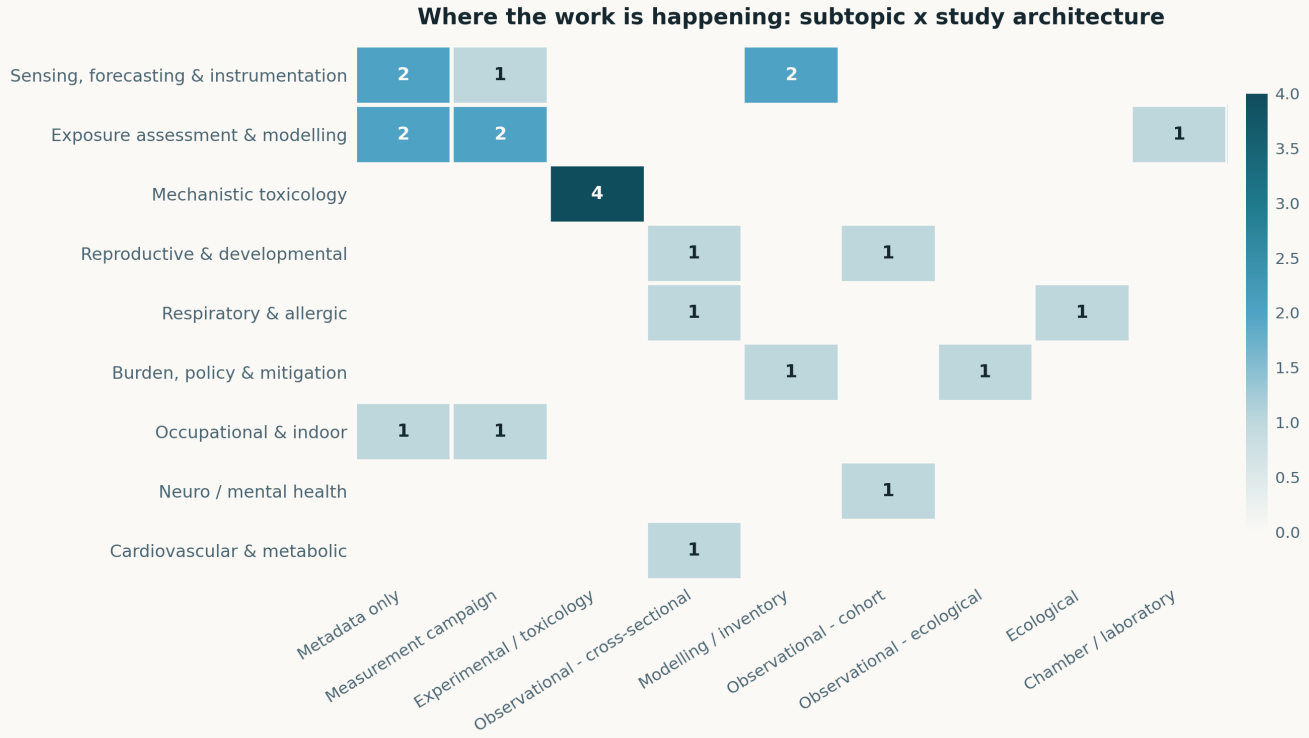


Fig. Subtopic × study architecture. Sensing splits between measurement, modelling and metadata cells; mechanistic toxicology is all experimental; the health bands sit in cohort, cross-sectional and ecological cells.

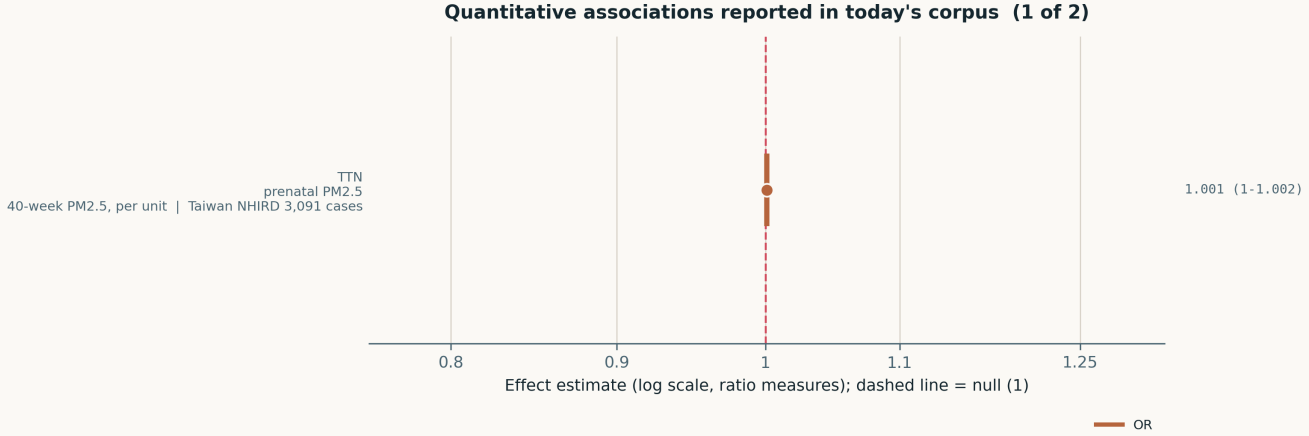


Fig. Ratio scale, one estimate: prenatal $PM_{2.5}$ and transient tachypnea of the newborn (aOR per unit concentration, Taiwan, 3,091 cases). The CI touches 1; CO, not $PM_{2.5}$, carries the study.

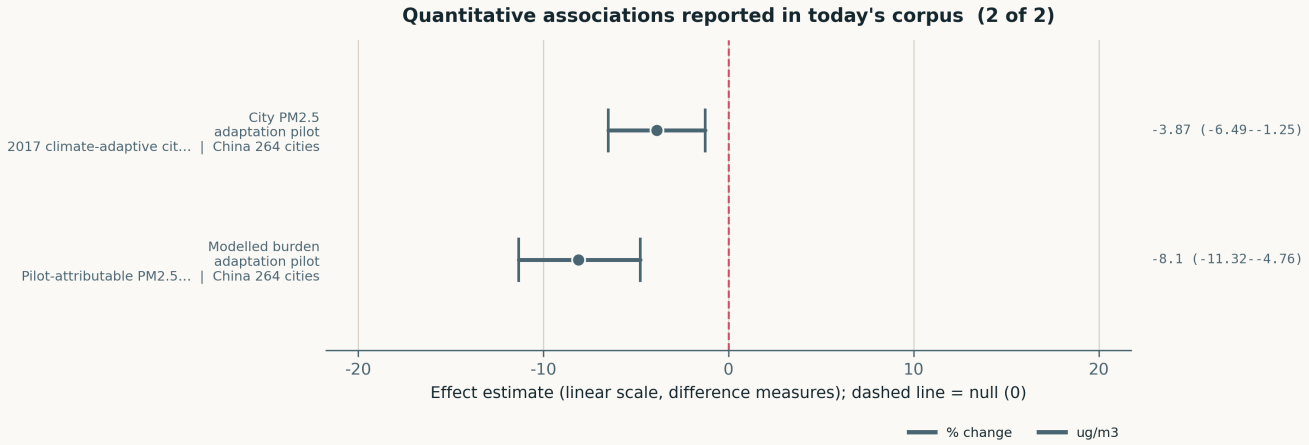


Fig. Difference scale, one study (China, 264 cities): the climate-adaptive city pilot's $PM_{2.5}$ change ($\mu g\,m^{-3}$) and the GEMM-modelled burden change (%). Different units; the second is derived from the first and is not an independent estimate.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

Yang heat anomaly wildfire smoke $PM_{2.5}$ incident depression All of Us 2026 J Affect Disord

Heat anomaly, wildfire smoke, social-material vulnerability, and incident depression in a national EHR-linked cohort: A preregistered secondary analysis

Preregistered Cox analysis in All of Us: 179,218 adults free of recorded depression, 5,484 incident cases over a mean 371 days. Exposures were the 30-day pre-index PRISM maximum-temperature anomaly, Stanford ECHO smoke $PM_{2.5}$ and a social-material vulnerability (SMV) composite. Heat anomaly: HR 1.06 (1.03-1.10) and 1.08 (1.04-1.11) per SD; SMV: HR 1.36 (1.33-1.40) per SD. Mean smoke $PM_{2.5}$ showed no association in the primary model; secondary smoke metrics were heterogeneous by metric and follow-up window. Limitation: exposure is assigned at the 3-digit ZIP level and frozen at baseline, so smoke-episode contrast is heavily smoothed and a null is weak evidence of no effect; EHR ascertainment depends on care access. Relevance: high - this is the misclassification problem that residence-scale sensor-corrected smoke $PM_{2.5}$ exists to fix.

doi: 10.1016/j.jad.2026.122663

Cardiovascular & metabolic — 1 record

Musa outdoor air pollution cardiovascular health school children Dar es Salaam 2026 BMC Public Health

Effects of outdoor air pollution on the cardiovascular health of school-aged children residing near polluted areas and the efficacy of mitigation measures in Dar es Salaam, Tanzania

Community cross-sectional study, December 2024-July 2025, of 653 school-aged children living in areas classed as highly polluted by proximity to industrial zones, major roads and waste-burning sites. Single oscillometric BP reading; 19.9% had elevated BP. In log-binomial models, prior respiratory disease carried ARR 8.81 (6.12-12.67) for cardiovascular symptoms. Awareness of mitigation policy was mixed. Limitation: no pollutant was measured and there is no unexposed comparison group, so the title's causal framing is not supported; one-visit BP overstates prevalence in children. Relevance: high as a gap marker - urban East Africa has almost no PM monitoring, and this design shows what health studies look like without it.

doi: 10.1186/s12889-026-28148-9

Respiratory & allergic — 2 records

Perez-Garcia nasal EWAS respiratory biomarkers Mexican American CHAMACOS 2026 medRxiv

Nasal Epigenome-Wide Association Study of Respiratory and Allergic Biomarkers in Mexican American Young Adults

Preprint. Nasal EWAS (EPICv2, 804,220 CpGs) in 354 CHAMACOS young adults aged 22, replicated in 540 Project Viva adolescents aged 13. Forty-nine CpGs were associated with FeNO and 557 DMRs with respiratory outcomes (FDR<0.05); 26 CpGs and 103 DMRs replicated (e.g. LRRFIP1, EPX, IL13). Ten FeNO-related CpGs were associated with prenatal and/or past-year $PM_{2.5}$ or O_3 at nominal p<0.05 only. Limitation: the air-pollution step is a secondary look-up without multiple-testing control, so the $PM_{2.5}$ link is hypothesis-generating; cross-sectional nasal sampling cannot order exposure and methylation. Relevance: low-medium - an agricultural-valley cohort where PM sources are local and poorly served by regulatory monitors.

doi: 10.64898/2026.10.27.26366640

Kobza respiratory ambulatory care air quality Upper Silesia 2026 Int J Occup Med Environ Health

The assessment of respiratory diseases in ambulatory care attributable to air quality in the Upper Silesian agglomeration

Five-year ecological analysis correlating daily mean concentrations from about 50 monitoring stations with National Health Fund outpatient incidence for seven respiratory ICD codes across eight regions of the Upper Silesian voivodeship. The strongest signal was for SO_2, not PM: tau 0.99 and rho 0.9 in regions 7 and 8, rho 0.78 in regions 1 and 4. Limitation: rank correlations on aggregated series without control for season, influenza, temperature or autocorrelation - seasonality alone would generate these coefficients, so the results carry no causal or PM-specific information. Relevance: low - included because $PM_{2.5}/PM_{10}$ were analysed; the coal-heating region is a classic case where denser residential PM sensing would sharpen exposure contrast.

doi: 10.13075/ijomeh.1896.02875

Reproductive & developmental — 2 records

Lin prenatal air pollution transient tachypnea newborn Taiwan 2026 Front Pediatr

Prenatal air pollution exposure and the risk of transient tachypnea of the newborn

Nationwide matched design on the Taiwan NHIRD linked to the Maternal and Child Health Database, 2004-2020: 3,091 TTN cases in first-born infants and 12,364 controls (1:4 on maternal age, sex, birth date). Exposure over the 40 weeks before birth from the nearest of 71 stations to the residential postal code. Adjusted ORs: CO 1.806 (1.312-2.487), NOx 1.010 (1.005-1.015), $PM_{2.5}$ 1.001 (1.000-1.002), the last per unit concentration with a CI touching the null. Effects concentrated in the Taipei basin. Limitation: nearest-station assignment at postal-code scale, no multipollutant model reported, and the $PM_{2.5}$ signal is marginal while CO dominates, pointing to traffic as a mixture rather than PM. Relevance: medium - traffic-gradient exposure in a dense basin is where street-level sensor networks change exposure contrast.

doi: 10.3389/fped.2026.1941575

Gaylord DNA methylation signature prenatal air pollution ECHO cohort 2026 Environ Epigenet

A methodology to identify a DNA methylation-based signature of prenatal air pollution exposure in the ECHO Cohort

Refines a pipeline that predicts high versus low in-utero air-pollution exposure from cord-blood DNA methylation (450K/EPIC) across multiple ECHO pregnancy cohorts. Exposures were ML-modelled at geocoded addresses; training and validation sets were matched on exposure level, site and array. Validation AUCs: $PM_{2.5}$ 0.64-0.68, NO_2 0.62-0.95, O_3 0.50-0.75, varying with region, exposure window and the high/low threshold. Limitation: the target is itself a model output, so the epigenetic classifier inherits exposure-model error, and $PM_{2.5}$ discrimination is modest; dichotomisation discards dose information. Relevance: medium - a biomarker of cumulative exposure is only as good as the exposure surface it is trained on, which is where denser monitoring would raise the ceiling.

doi: 10.1093/eep/dvag034

Mechanistic toxicology 4 records

He FOXM1 lncRNA COPD-R biomass $PM_{2.5}$ pulmonary inflammation rat 2026 Ecotoxicol Environ Saf

FOXM1/LncRNA COPD-R regulates pulmonary inflammation in chronic obstructive pulmonary disease induced by biomass-related $PM_{2.5}$

Rats exposed to biomass-fuel smoke for seven months developed COPD-like disease; lung RNA-seq found 312 differentially expressed lncRNAs (294 up). lncRNA COPD-R rose from 0.15 to 1.05 (relative expression, $p<0.01$). In rat L2 alveolar cells, biomass-related $PM_{2.5}$ increased COPD-R; knockdown reduced PM-induced CCL2, IL-17F, MMP-9 and TIMP-1, and ChIP/luciferase assays placed FOXM1 upstream as its transcriptional activator, with rescue experiments consistent with a FOXM1-COPD-R axis. Limitation: whole-smoke exposure in vivo but $PM_{2.5}$ only in vitro, dose not reported in the abstract, and a rat lncRNA whose human orthologue is not established. Relevance: low - mechanistic support for household biomass PM, where the exposure gap is measurement in homes.

doi: 10.1016/j.ecoenv.2026.120952

Chen pirfenidone fine particulate matter EMT mitochondrial fission autophagy 2026 Respir Res

Pirfenidone reduces fine particulate matter-induced epithelial to mesenchymal transition in lung epithelial cells through regulating mitochondrial fission and autophagy

A549 cells exposed to 100 ug/mL fine PM lost E-cadherin and gained vimentin, alpha-SMA, TGF-beta1, Snail and Slug, turned spindle-shaped and migrated more; PM increased mitochondrial fission and LC3B/p62 accumulation, i.e. impaired autophagic flux. Intratracheal PM in mice reproduced EMT markers, inflammatory infiltration and alveolar wall thickening. Pirfenidone reversed these effects in vitro and reduced collagen deposition in vivo. Limitation: 100 ug/mL on a cell monolayer and bolus instillation are far above realistic deposited doses, and A549 is a carcinoma line; the drug result says nothing about population exposure. Relevance: low - mechanism for PM-associated fibrosis; no measurement content.

doi: 10.1186/s12931-026-03804-1

Zakari PM_1 microbial profiles lung gut microbiome mice 2026 J Hazard Mater

Associations of particulate matter-associated microbial profiles with lung and intestinal microbiomes in healthy mice

Mice were housed under ambient PM_1, HEPA-filtered air or clean air; PM_1 was characterised for ions, metals, carbon fractions, oxidative potential and its own 16S microbiome (Proteobacteria-dominated). Lung diversity barely changed while the gut microbiome shifted: PM_1 reduced Actinobacteriota and Bacteroidota and raised Proteobacteria versus both control arms. Eleven families and six genera overlapped among PM_1, lung and gut communities in the PM_1 group. Limitation: group sizes and exposure concentrations are not given in the abstract, gases co-vary with PM_1, and taxon overlap does not show transfer from air to gut. Relevance: low - airborne microbial load is a PM component no optical sensor resolves.

doi: 10.1016/j.jhazmat.2026.143978

Gautam diesel exhaust miR-133b-3p CTGF Hippo ECM remodeling mouse lung 2026 Environ Mol Mutagen

Diesel Exhaust-Induced miR-133b-3p Downregulation and CTGF-Linked Hippo Pathway Inactivation Promote ECM Remodeling and Inflammation in Mouse Lungs

C57BL/6 mice received diesel exhaust 30 min/day, 5 days/week for 8 weeks versus filtered air. Microarrays showed miR-133b-3p down more than 2-fold; CTGF was confirmed as a direct target in A549 cells (mimic reduced CTGF mRNA $p=0.0054$, protein $p=0.0001$). In vivo, exhaust raised CTGF, lowered P-YAP/YAP (Hippo inactivation), increased BALF IL-18, IL-1beta and GM-CSF, raised MMP-3/7/12 and lowered LOXL1/4, consistent with ECM remodelling. Limitation: exhaust dose (PM mass, NO_2) is not reported in the abstract, so particle and gas effects cannot be separated; the CTGF link is correlative in vivo. Relevance: low - traffic-exhaust mechanism; supports particle number or black carbon over mass as the near-road metric.

doi: 10.1002/em.70088

Exposure assessment & modelling 5 records

Yousaf nanoscale curvature hygroscopic growth respiratory deposition ultrafine 2026 Environ Pollut

Nanoscale Curvature Controls Hygroscopic Growth and Respiratory Deposition of Atmospheric Nanoparticles

HTDMA measurements of 15-100 nm NaCl particles fed into the MPPD lung-deposition model. The Kelvin effect raised deliquescence RH to 77-81% for 15-36 nm particles and suppressed growth; at 80-90% RH, growth factors were 1.95-2.25 (15 nm) and 1.99-2.36 (100 nm). Ignoring hygroscopic growth overestimated total deposition by up to 25%, with an 18% tracheobronchial difference for particles <=50 nm, and shifted larger UFPs deeper into the lung. Limitation: pure NaCl is a best-case hygroscopic aerosol; ambient UFP is organic-rich with far lower kappa, so the correction is an upper bound. Relevance: medium - the same size-dependent water uptake that biases optical sensors in humid air also changes delivered dose, so dry-size number counts understate the problem.

doi: 10.1016/j.envpol.2026.129429

Kalkavouras ammonia sulfuric acid BVOC urban new particle formation Athens 2026 Environ Pollut

Assessing the combined roles of ammonia, sulfuric acid, and biogenic volatile organic compounds in urban new particle formation

Seven months (June 2023-January 2024) of particle number size distributions, NH_3 and BVOCs at the Thissio urban background site in Athens, with a proxy for H2SO4. Mean NH_3 2.3 +/- 0.7 $\mu g\,m^{-3}$. NPF occurred on 19% of days, most often in autumn (29-33%) when ventilation and condensation sink were low. Precursor levels barely differed between event and non-event days; the highest J14 came at intermediate NH_3 and H2SO4 with low BVOCs, while growth rates rose with NH_3 and isoprene/monoterpenes. Limitation: H2SO4 is a proxy, not measured, and the record lacks a full annual cycle. Relevance: medium - NPF bursts dominate urban number concentration but are invisible to mass-based optical sensors, which see nothing below ~300 nm.

doi: 10.1016/j.envpol.2026.129453

Chakraborty fog intensity droplet size aqueous SOA Kanpur 2026 Environ Sci Technol

Fog Intensity and Droplet Size-Resolved Aqueous Secondary Organic Aerosol (aqSOA) Formation in Polluted Winter Fog: New Insights from a Combined Observational and Modeling Study

Two winters of size-resolved fogwater at Kanpur analysed by offline HR-AMS and PMF; three highly oxidised factors (O/C 0.95-1.19) were summed as an aqSOA proxy and scaled by liquid water content and WSOC. Across seven events aqSOA was 3.1-8.4 $\mu g\,m^{-3}$, a median ~7% of ambient SOA and 5% of OA, up to 13% of SOA in the densest fogs. Fine droplets carried 43-84% of aqSOA, linked to dissolved Fe and Cu and longer residence. Limitation: the aqSOA proxy is a PMF construct, and offline AMS of fogwater may not represent residual particle mass after evaporation. Relevance: medium - Indo-Gangetic fog episodes are when optical PM sensors fail outright from droplet interference, while secondary mass is being added.

doi: 10.1021/acs.est.6c09700

Yingkar $PM_{2.5}$-bound PAHs Central Asia urban trends sources risk 2026 Atmos Environ

Environment-to-Health Linkages of $PM_{2.5}$-Bound Polycyclic Aromatic Hydrocarbons in Urban Cities of Central Asia: Trends, Sources, and Risk Assessment

METADATA ONLY. Title and authors only (Yingkar, Omarova, Sovetova, Alibekov et al.). No abstract was retrievable from Crossref or Semantic Scholar at harvest. The title describes trends, source apportionment and health-risk assessment of $PM_{2.5}$-bound PAHs across urban Central Asian cities - a region with sparse speciation data and heavy coal heating. Cities, sampling design, concentrations and risk estimates are not available; nothing is inferred. Relevance: medium - coal-heating PAH peaks in Almaty-type basins are where low-cost PM networks already operate without any chemical reference; retry the abstract on a later run.

doi: 10.1016/j.atmosenv.2026.122447

Fang end of uniformity spatial climatic toxicological redefinitions air quality 2026 Environ Sci Technol

The End of Uniformity: Spatial, Climatic, and Toxicological Redefinitions of Air Quality

METADATA ONLY. Title and authors only (Fang, Song, Weon). No abstract was retrievable at harvest; the format is likely a viewpoint or perspective. The title argues against uniform air-quality definitions along three axes - spatial heterogeneity, climate-driven change and toxicity beyond mass - which maps directly onto hyperlocal sensing, wildfire-era exposure and component- or oxidative-potential-based metrics. Arguments and any proposals are not available; nothing is inferred. Relevance: medium - if it advocates spatially resolved or toxicity-weighted standards, it is a policy argument for distributed networks; retry the text on a later run.

doi: 10.1021/acs.est.6c14170

Sensing, forecasting & instrumentation 5 records

Waghmare machine learning explainable AI $PM_{2.5}$ prediction Nagpur 2026 MethodsX

Machine learning models with explainable AI for $PM_{2.5}$ prediction for Urban air quality assessment

Six regressors (decision tree, extra trees, random forest, KNN, XGBoost, SVR) trained on ~4,000 observations of $PM_{2.5}$, NO_2, SO_2, NH_3, O_3 and meteorology in Nagpur, 70/30 split with 10-fold CV. A single decision tree scored best: test R2 0.90, RMSE 7.5-8.9 $\mu g\,m^{-3}$, MAE 5.4-6.6 $\mu g\,m^{-3}$; SHAP ranked NO_2, O_3, RH and NH_3 top. Limitation: random splits of autocorrelated time series leak information, so skill is inflated; co-pollutants measured at the same station make this a nowcast, not a forecast, and a lone tree beating ensembles is a warning sign. Relevance: low-medium - illustrates the validation discipline (blocked or spatial CV) that sensor-network ML calibration needs.

doi: 10.1016/j.mex.2026.104217

Wu source-dependent organic aerosol hygroscopicity eastern China biomass burning 2026 Atmos Chem Phys

Source-dependent organic aerosol hygroscopicity in eastern China: observational evidence for conditional enhancement under biomass-burning influence

Winter field campaign at a rural eastern-China site retrieving organic-aerosol hygroscopicity from humidified nephelometry and ZSR closure. kappa_OA ranged from ~0 to 0.49 (mean 0.11 +/- 0.11) and correlated only weakly with O:C; instead it rose with biomass-burning influence, and ageing enhanced kappa_OA only when biomass burning dominated. Highly oxidised OA under weak biomass influence stayed weakly hygroscopic. Limitation: kappa_OA is a residual after subtracting inorganic contributions, so errors in the inorganic closure load onto the organic term; one site, one season. Relevance: high - humidity correction of optical PM sensors usually assumes a fixed kappa; this shows kappa shifts with source, so smoke-affected periods need a different growth correction.

doi: 10.5194/acp-26-15333-2026

Schmidt AOD gap filling iterative singular spectrum analysis AERONET Cuiaba 2026 Environ Monit Assess

Gap filling in long-term AOD time series using Iterative Singular Spectrum Analysis at a tropical AERONET site

Applies iterative singular spectrum analysis (ISSA) to fill a 2005-2024 daily AOD record from the Cuiaba-Miranda AERONET station, where ~47% of days are missing, mostly in the cloudy wet season. Climatological (day-of-year mean) initialisation beat global-mean initialisation, giving lower error and better temporal consistency, and the cross-validation design materially changed the optimal reconstruction parameters. No error magnitudes are given in the abstract. Limitation: gaps are not missing at random - they coincide with cloud and wet-season aerosol regimes - so a climatology-anchored fill will bias toward dry-season behaviour exactly where data are absent. Relevance: low-medium - the same informative-missingness problem affects sensor networks with humidity-triggered dropouts and satellite-surface fusion.

doi: 10.1007/s10661-026-16067-6

Trivanovic periodic technical inspection particle number devices gasoline exhaust 2026 J Aerosol Sci

Assessment of existing PN-PTI devices to measure gasoline exhaust

METADATA ONLY. Title and authors only (Trivanovic, Melas, Auderset, Hammer et al.). No abstract was retrievable at harvest. The title indicates an evaluation of existing particle-number periodic-technical-inspection (PN-PTI) instruments - developed and type-approved for diesel vehicles with particle filters - when applied to gasoline exhaust, whose size distribution sits lower and nearer the counters' 23 nm or 10 nm cut. Instruments, test vehicles and error magnitudes are not available; nothing is inferred. Relevance: high - PN-PTI counters are the first mass-produced, low-cost, legally metrological particle counters, and their size-cut behaviour is the same question for community UFP sensing.

doi: 10.1016/j.jaerosci.2026.106901

Spinosa volcanic Saharan dust biomass burning aerosol advections Naples 2026 Atmos Res

Multiple advections of volcanic, Saharan dust and biomass burning aerosols over Naples

METADATA ONLY. Title and authors only (Spinosa, Manzo, Amoruso, Sannino et al.). No abstract was retrievable at harvest. The title describes overlapping advections of volcanic, Saharan dust and biomass-burning aerosol over Naples; the author group operates lidar there, so a profiling case study is likely but is not confirmed. Instruments, dates and surface PM impacts are not available; nothing is inferred. Relevance: medium - mixed dust/smoke/volcanic layers are the cases where optical low-cost sensors misreport mass most, because refractive index and size change simultaneously.

doi: 10.1016/j.atmosres.2026.109425

Occupational & indoor 2 records

Martin simplified method portable air cleaner CADR low-cost sensors 2026 MethodsX

A simplified method for evaluating fine particulate matter removal performance of portable air cleaners

Methods paper for community-run CADR testing: release a short-lived nebuliser aerosol in a sealed room, log $PM_{2.5}$ decay with and without the cleaner on low-cost sensors, and compute CADR from the decay-rate difference. Residential low-cost-sensor CADRs ran ~26% below laboratory values from research-grade instruments - a conservative bias. Of ten units, a four-filter Corsi-Rosenthal box performed best among DIY builds; a fan shroud added ~11%. Across 135 models DIY units cost far less per CADR unit. Limitation: the 26% gap mixes sensor size-response with room mixing and leakage, and is not decomposed; nebuliser aerosol is not wildfire smoke. Relevance: high - a practical protocol in which the low-cost sensor is the reference, with its bias quantified.

doi: 10.1016/j.mex.2026.104216

Hunter e-cigarette aerosol early post-emission particle decay controlled conditions 2026 Sci Total Environ

Early post-emission particle decay dynamics of E-cigarette aerosols under controlled conditions

METADATA ONLY. Title and authors only (Hunter, Jacoby, Ferdousi-Rokib, Asa-Awuku et al.). No abstract was retrievable at harvest. The title describes the first seconds-to-minutes decay of e-cigarette aerosol under controlled conditions - evaporation of volatile droplets that shrinks particles rapidly after exhalation or emission. Instruments, conditions and decay constants are not available; nothing is inferred. Relevance: medium - e-cigarette droplets produce large, short-lived spikes on indoor optical PM sensors that mass-based readings misstate because the particles are liquid and evaporating; useful for indoor network event classification.

doi: 10.1016/j.scitotenv.2026.182379

Burden, policy & mitigation 2 records

Wang climate-adaptive city pilot $PM_{2.5}$ health dividend China 2026 Front Public Health

Does climate adaptation policy deliver a health dividend? Exposure, modeled health burden, and defensive capacity in China's climate-adaptive city pilots

Difference-in-differences on a 2011-2022 panel of 264 prefecture cities, 21 treated by China's 2017 climate-adaptive city pilot. The pilot is associated with $PM_{2.5}$ lower by 3.87 $\mu g\,m^{-3}$ (95% CI -6.49 to -1.25; wild-bootstrap p=0.0067); GEMM translates this into an ~8.1% lower modelled burden (-11.32 to -4.76%) - an accounting result, not observed mortality. Defensive-capacity responses (physicians, beds, insurance) are imprecise. Limitation: the estimate shrinks and loses precision in the matched sample and after pollution-convergence adjustment, so selection of already-improving cities is not excluded; 21 treated units is a small base. Relevance: low-medium - policy evaluation at city scale rests on gridded $PM_{2.5}$ whose own error is not propagated.

doi: 10.3389/fpubh.2026.1987084

Wang carbon footprint air-pollutant emissions Beijing 2022 Winter Olympics 2026 Atmosphere

Linking Carbon Footprints to Air-Pollutant Emissions from Mega Sporting Events: Evidence from the Beijing 2022 Winter Olympics

Screening-level inventory converting the verified 713,788 t CO_2e footprint of the Beijing 2022 Winter Olympics (2016-2022) into co-emitted pollutants via Chinese pollutant-to-CO_2 ratios over eight emission carriers, with Shapley attribution and Monte Carlo/Sobol uncertainty. Construction produced 63% of CO_2e, ~70% of 1,335 t NOx and 76% of 279 t $PM_{2.5}$. Green electricity avoided 31% of no-measure emissions; offsets reduced no air pollutant. Limitation: emission-side only - ratio-based $PM_{2.5}$ with no dispersion, concentration or exposure step, so there is no health quantity, and national average ratios may misstate construction-site fugitive dust. Relevance: low - construction dust is the source class that site-perimeter low-cost PM monitoring is typically deployed to police.

doi: 10.3390/atmos17111072

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Yang heat anomaly wildfire smoke $PM_{2.5}$ incident depression All of Us	Heat anomaly, wildfire smoke, social-material vulnerability, and incident depression in a national EHR-linked cohort: A preregistered secondary analysis	*J Affect Disord*	Prospective cohort (registry linkage)	30-day mean wildfire-smoke $PM_{2.5}$ (Stanford ECHO model) at ZIP3	USA (national, All of Us)
Cardiovascular & metabolic					
Musa outdoor air pollution cardiovascular health school children Dar es Salaam	Effects of outdoor air pollution on the cardiovascular health of school-aged children residing near polluted areas and the efficacy of mitigation measures in Dar es Salaam, Tanzania	*BMC Public Health*	Cross-sectional survey	None measured; residence near industry, major roads or waste burning used as proxy	Tanzania (Dar es Salaam)
Reproductive & developmental					
Gaylord DNA methylation signature prenatal air pollution ECHO cohort	A methodology to identify a DNA methylation-based signature of prenatal air pollution exposure in the ECHO Cohort	*Environ Epigenet*	Cross-sectional (birth cohort)	Prenatal $PM_{2.5}$, NO2, 8-h max O3 from geocoded ML exposure models	USA (multi-site ECHO cohorts)
Lin prenatal air pollution transient tachypnea newborn Taiwan	Prenatal air pollution exposure and the risk of transient tachypnea of the newborn	*Front Pediatr*	Case-cohort study	40-week prenatal $PM_{2.5}$, CO, NOx from nearest of 71 monitoring stations (postal code)	Taiwan (national)
Respiratory & allergic					
Kobza respiratory ambulatory care air quality Upper Silesia	The assessment of respiratory diseases in ambulatory care attributable to air quality in the Upper Silesian agglomeration	*Int J Occup Med Environ Health*	Ecological	Daily $PM_{2.5}$, PM_{10}, SO2, NOx, Pb, benzene from 50 regional stations	Poland (Upper Silesia)
Perez-Garcia nasal EWAS respiratory biomarkers Mexican American CHAMACOS	Nasal Epigenome-Wide Association Study of Respiratory and Allergic Biomarkers in Mexican American Young Adults	*medRxiv*	Cross-sectional (cohort baseline)	Prenatal and past-year $PM_{2.5}$ and O3 (residential)	USA (Salinas Valley, California, CHAMACOS; Project Viva replication)
Mechanistic toxicology					
Chen pirfenidone fine particulate matter EMT mitochondrial fission autophagy	Pirfenidone reduces fine particulate matter-induced epithelial to mesenchymal transition in lung epithelial cells through regulating mitochondrial fission and autophagy	*Respir Res*	Animal + in vitro	Fine PM (standard reference material type not stated), 100 ug/mL in vitro; intratracheal in mice	Laboratory (no field site)
Gautam diesel exhaust miR-133b-3p CTGF Hippo ECM remodeling mouse lung	Diesel Exhaust-Induced miR-133b-3p Downregulation and CTGF-Linked Hippo Pathway Inactivation Promote ECM Remodeling and Inflammation in Mouse Lungs	*Environ Mol Mutagen*	Animal exposure study	Whole diesel exhaust, 30 min/day, 5 d/week, 8 weeks vs filtered air	India (inferred from author group; laboratory)
He FOXM1 lncRNA COPD-R biomass $PM_{2.5}$ pulmonary inflammation rat	FOXM1/LncRNA COPD-R regulates pulmonary inflammation in chronic obstructive pulmonary disease induced by biomass-related $PM_{2.5}$	*Ecotoxicol Environ Saf*	Animal + in vitro	Biomass-fuel smoke (rats, 7 months) and biomass-related $PM_{2.5}$ (L2 cells)	China (inferred from author group; laboratory)
Zakari PM1 microbial profiles lung gut microbiome mice	Associations of particulate matter-associated microbial profiles with lung and intestinal microbiomes in healthy mice	*J Hazard Mater*	Murine inhalation exposure	Ambient PM1 (ions, metals, carbon, oxidative potential, 16S) vs HEPA-filtered and clean air	Not stated in record (ambient-air exposure chamber)
Occupational & indoor					
Hunter e-cigarette aerosol early post-emission particle decay controlled conditions	Early post-emission particle decay dynamics of E-cigarette aerosols under controlled conditions	*Sci Total Environ*	Metadata only (no abstract)	E-cigarette aerosol particle size/number decay	Not stated in record
Martin simplified method portable air cleaner CADR low-cost sensors	A simplified method for evaluating fine particulate matter removal performance of portable air cleaners	*MethodsX*	Field co-location + chamber	$PM_{2.5}$ decay measured with low-cost sensors vs research-grade instruments	USA (residence and laboratory)

First author	Focus	Journal	Design	Metric	Region
Sensing, forecasting & instrumentation					
Schmidt AOD gap filling iterative singular spectrum analysis AERONET Cuiaba	Gap filling in long-term AOD time series using Iterative Singular Spectrum Analysis at a tropical AERONET site	*Environ Monit Assess*	Method / algorithm	AERONET daily aerosol optical depth	Brazil (Cuiaba-Miranda, Mato Grosso)
Spinosa volcanic Saharan dust biomass burning aerosol advections Naples	Multiple advections of volcanic, Saharan dust and biomass burning aerosols over Naples	*Atmos Res*	Metadata only (no abstract)	Columnar/ profiled aerosol type (likely lidar-based)	Italy (Naples)
Trivanovic periodic technical inspection particle number devices gasoline exhaust	Assessment of existing PN-PTI devices to measure gasoline exhaust	*J Aerosol Sci*	Metadata only (no abstract)	Solid particle number (PN) at tailpipe, PTI-grade counters	Switzerland (inferred from author group; not stated in record)
Waghmare machine learning explainable AI $PM_{2.5}$ prediction Nagpur	Machine learning models with explainable AI for $PM_{2.5}$ prediction for Urban air quality assessment	*MethodsX*	Gradient-boosting model + SHAP interpretability	Station $PM_{2.5}$ with co-pollutants and meteorology	India (Nagpur, Maharashtra)
Wu source-dependent organic aerosol hygroscopicity eastern China biomass burning	Source-dependent organic aerosol hygroscopicity in eastern China: observational evidence for conditional enhancement under biomass-burning influence	*Atmos Chem Phys*	Field campaign	Humidified light scattering f(RH) with OA composition; ZSR closure for kappa_OA	China (eastern, rural site)
Exposure assessment & modelling					
Chakraborty fog intensity droplet size aqueous SOA Kanpur	Fog Intensity and Droplet Size-Resolved Aqueous Secondary Organic Aerosol (aqSOA) Formation in Polluted Winter Fog: New Insights from a Combined Observational and Modeling Study	*Environ Sci Technol*	Field campaign	Size-resolved fogwater, offline HR-AMS with PMF; ambient OA	India (Kanpur)
Fang end of uniformity spatial climatic toxicological redefinitions air quality	The End of Uniformity: Spatial, Climatic, and Toxicological Redefinitions of Air Quality	*Environ Sci Technol*	Metadata only (no abstract)	Not stated (perspective on air-quality metrics)	Not stated in record
Kalkavouras ammonia sulfuric acid BVOC urban new particle formation Athens	Assessing the combined roles of ammonia, sulfuric acid, and biogenic volatile organic compounds in urban new particle formation	*Environ Pollut*	Long-term field measurement campaign	Particle number size distributions with NH3, BVOCs and proxy H2SO4	Greece (Athens, Thissio)
Yingkar $PM_{2.5}$-bound PAHs Central Asia urban trends sources risk	Environment-to-Health Linkages of $PM_{2.5}$-Bound Polycyclic Aromatic Hydrocarbons in Urban Cities of Central Asia: Trends, Sources, and Risk Assessment	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$-bound PAHs	Kazakhstan (Central Asian cities; inferred from title and author group)
Yousaf nanoscale curvature hygroscopic growth respiratory deposition ultrafine	Nanoscale Curvature Controls Hygroscopic Growth and Respiratory Deposition of Atmospheric Nanoparticles	*Environ Pollut*	Laboratory measurement	15-100 nm NaCl particles, HTDMA growth factors fed to MPPD dosimetry	Laboratory (no field site)
Burden, policy & mitigation					
Wang carbon footprint air-pollutant emissions Beijing 2022 Winter Olympics	Linking Carbon Footprints to Air-Pollutant Emissions from Mega Sporting Events: Evidence from the Beijing 2022 Winter Olympics	*Atmosphere*	Emission inventory	Co-emitted $PM_{2.5}$, NOx, SO2, CO, NMVOC from pollutant-to-CO2 ratios	China (Beijing)
Wang climate-adaptive city pilot $PM_{2.5}$ health dividend China	Does climate adaptation policy deliver a health dividend? Exposure, modeled health burden, and defensive capacity in China's climate-adaptive city pilots	*Front Public Health*	Difference-in-differences evaluation	City-level annual $PM_{2.5}$ (gridded product), GEMM burden	China (264 cities)

Provenance and method

Entry window **30 October 2026**, single day (Friday). Harvester legs: **PubMed** health **16**, sensing **4**; **Europe PMC 40** (9 AGRICOLA retro-index records dropped); **Crossref by ISSN 43**; **arXiv** 150 kB, no in-window entry; **OpenAlex** HTTP 429. 95 unique, 10 already seen, 85 fresh; PubMed connector (EDAT 30–31 Oct, health and sensing axes separately) 27 PMIDs, 18 bucketed to 30 Oct, 1 merged, **6 appended**, **91 screened**. Abstracts for 22 PubMed records recovered by efetch; Crossref/Semantic Scholar recovered further abstracts, leaving 19 candidates abstract-free. **Consensus** calibration sweep (once for 30–31 October): 10 hits, none new for this day. **24 admitted**, **67 rejected** with logged reasons (5 were EPMC re-surfacings of 29 October rejections; 1 duplicate MethodsX record). **Trial watch** 30 October–1 November: 0 registrations updated. **5 carried on metadata alone**. **Scholar Gateway** not called. Abstracts remain the copyright of their publishers.